

Calculus III — Chain Rule and Directional Derivatives (Problems 18–19, 23–26)

Chain Rule. For $F(s, t) = f(x(s, t), y(s, t))$:

$$F_s = f_x x_s + f_y y_s, \quad F_t = f_x x_t + f_y y_t$$

(same pattern for $f(r, \theta)$).

Problem 18

a) $f = x^2 y^3$, $x = s \cos t$, $y = s \sin t$

$$F_s = 2xy^3 \cos t + 3x^2 y^2 \sin t = 5s^4 \cos^2 t \sin^3 t$$

$$F_t = -2sxy^3 \sin t + 3sx^2 y^2 \cos t = s^5 \sin^2 t \cos t (3 \cos^2 t - 2 \sin^2 t)$$

b) $f = \arcsin(x - y)$, $x = s^2 + t^2$, $y = 1 - 2st$

$$F_s = \frac{2s + 2t}{\sqrt{1 - (x - y)^2}}, \quad F_t = \frac{2t + 2s}{\sqrt{1 - (x - y)^2}}$$

Since $x - y = (s + t)^2 - 1$, both simplify to

$$F_s = F_t = \frac{2(s + t)}{|s + t| \sqrt{2 - (s + t)^2}}$$

c) $f = e^{x+2y}$, $x = s/t$, $y = t/s$

$$F_s = e^{s/t+2t/s} \left(\frac{1}{t} - \frac{2t}{s^2} \right), \quad F_t = e^{s/t+2t/s} \left(-\frac{s}{t^2} + \frac{2}{s} \right)$$

d) $f = \tan(x/y)$, $x = 2s + 3t$, $y = 3s - 2t$

$$F_s = \frac{-13t}{(3s - 2t)^2} \sec^2 \left(\frac{2s + 3t}{3s - 2t} \right), \quad F_t = \frac{13s}{(3s - 2t)^2} \sec^2 \left(\frac{2s + 3t}{3s - 2t} \right)$$

Problem 19

a) $f(r, \theta) = \sin \theta \cos r$, $r = st^2$, $\theta = s^2t$

$$F_s = -t^2 \sin(s^2t) \sin(st^2) + 2st \cos(s^2t) \cos(st^2)$$

$$F_t = -2st \sin(s^2t) \sin(st^2) + s^2 \cos(s^2t) \cos(st^2)$$

b) $f(r, \theta) = e^r \cos \theta$, $r = st$, $\theta = \sqrt{s^2 + t^2}$

$$F_s = e^{st} \left[t \cos \sqrt{s^2 + t^2} - \frac{s}{\sqrt{s^2 + t^2}} \sin \sqrt{s^2 + t^2} \right]$$

$$F_t = e^{st} \left[s \cos \sqrt{s^2 + t^2} - \frac{t}{\sqrt{s^2 + t^2}} \sin \sqrt{s^2 + t^2} \right]$$

Directional Derivative. The directional derivative of a differentiable function f in the direction of a unit vector $\mathbf{u}$ represents the rate of change of the function in that direction, given by:

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$$

where ∇f is the gradient vector of f :

- In 2D: $\nabla f(x, y) = \langle f_x, f_y \rangle = f_x \mathbf{i} + f_y \mathbf{j}$
- In 3D: $\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k}$

Finding the Unit Vector $\mathbf{u}$: The direction vector must be normalized to a unit vector (length 1) before computing the directional derivative.

1. **Given an angle θ (in 2D):** The unit vector is

$$\mathbf{u} = \langle \cos \theta, \sin \theta \rangle$$

2. **Given a direction vector $\mathbf{v}$:** Normalize $\mathbf{v}$:

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

3. **Given a direction from point P to point Q :** The displacement vector is $\mathbf{v} = \overrightarrow{PQ} = Q - P$, and the unit vector is:

$$\mathbf{u} = \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|}$$

Problem 23

Find the directional derivative of f at the given point in the direction indicated by the angle θ .

- a) $f(x, y) = xy^3 - x^2$, $(1, 2)$, $\theta = \pi/3$.
- b) $f(x, y) = y \cos(xy)$, $(0, 1)$, $\theta = \pi/4$.
- c) $f(x, y) = \sqrt{2x + 3y}$, $(3, 1)$, $\theta = -\pi/6$.

Problem 24

Find the directional derivative of the function at the given point in the direction of the vector $\mathbf{v}$.

- a) $f(x, y) = e^x \sin y$, $(0, \pi/3)$, $\mathbf{v} = \langle -6, 8 \rangle$.
- b) $f(x, y) = \frac{x}{x^2 + y^2}$, $(1, 2)$, $\mathbf{v} = \langle 3, 5 \rangle$.
- c) $g(s, t) = s\sqrt{t}$, $(2, 4)$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$.
- d) $g(u, v) = u^2 e^{-v}$, $(3, 0)$, $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$.
- e) $f(x, y, z) = x^2 y + y^2 z$, $(1, 2, 3)$, $\mathbf{v} = \langle 2, -1, 2 \rangle$.

Problem 25

- a) Find the directional derivative of $f(x, y, z) = xy^2z^3$ at $P(2, 1, 1)$ in the direction of $Q(0, -3, 5)$.
b) Find the directional derivative of $f(x, y) = \sqrt{xy}$ at $P(2, 8)$ in the direction of $Q(5, 4)$.

Problem 26

Find the rate of change of f at P in the direction of the vector $\mathbf{u}$.

- a) $f(x, y) = x/y$, $P(2, 1)$, $\mathbf{u} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$.
b) $f(x, y) = x^2 \ln y$, $P(3, 1)$, $\mathbf{u} = -\frac{5}{13}\mathbf{i} + \frac{12}{13}\mathbf{j}$.
c) $f(x, y, z) = x^2yz - xyz^3$, $P(2, -1, 1)$, $\mathbf{u} = \langle 0, \frac{4}{5}, -\frac{3}{5} \rangle$.
d) $f(x, y, z) = y^2e^{xyz}$, $P(0, 1, -1)$, $\mathbf{u} = \langle \frac{3}{13}, \frac{4}{13}, \frac{12}{13} \rangle$.